

Advanced SQL

Assignment Outputs

Question 1: What is a CTE, and how does it improve readability?

A Common Table Expression (CTE) is a named, temporary result set defined using a WITH clause that exists only for the duration of a single query. It improves readability by breaking a complex query into logical, named steps, avoiding repeated subqueries, and making multi-step logic (like aggregations followed by filtering) easier to follow and maintain.

Question 2: Why are some views updatable while others are read-only?

A view is updatable when the database can map each row of the view back to exactly one row in a single underlying base table, with no aggregation, GROUP BY, DISTINCT, joins across multiple tables, or computed/derived columns involved. A view is read-only when it aggregates data (e.g. COUNT, AVG), joins multiple tables, or uses set operations, because changes can no longer be unambiguously traced back to a single source row.

Example: a view selecting ProductID, ProductName, Price directly from the Products table is updatable. A view like vw_CategorySummary, which uses GROUP BY and AVG(Price), is read-only.

Question 3: Advantages of stored procedures over raw SQL queries

Stored procedures improve performance since they are precompiled and cached by the database engine, reduce network traffic by executing multiple statements server-side in one call, promote code reuse and consistency by centralizing logic instead of duplicating it across applications, improve security by allowing controlled access without exposing underlying tables directly, and simplify maintenance since logic changes only need to be updated in one place.

Question 4: Purpose of triggers and an essential use case

A trigger is a block of SQL code that automatically executes in response to a specific event (INSERT, UPDATE, or DELETE) on a table, enforcing business rules or automating side-effect actions without requiring the application to explicitly call them.

Essential use case: automatically archiving deleted records. An AFTER DELETE trigger on the Products table can copy each deleted row into a ProductArchive table, preserving historical data even after the original row is removed, which is critical for audit trails.

Question 5: Need for data modelling and normalization

Data modelling defines how entities, attributes, and relationships are structured before a database is built, ensuring the schema accurately reflects real-world business rules and supports the queries the application will need. Normalization organizes this schema to eliminate redundant data and avoid update, insert, and delete anomalies, by splitting data into related tables connected via keys. Together they reduce storage waste, keep

data consistent across the database, and make the schema easier to scale and maintain as requirements grow.

Question 6: Total revenue per product (CTE), revenue > 3000

ProductID	ProductName	Price	Quantity	Revenue
1	Keyboard	1200.0	4	4800.0
2	Mouse	800.0	10	8000.0
3	Chair	2500.0	2	5000.0
4	Desk	5500.0	1	5500.0

All 4 products qualify since every product's revenue exceeds 3000.

Question 7: vw_CategorySummary (Category, TotalProducts, AveragePrice)

Category	TotalProducts	AveragePrice
Electronics	2	1000.0
Furniture	2	4000.0

Question 8: Updatable view (ProductID, ProductName, Price)

View contents before updating ProductID = 1:

ProductID	ProductName	Price
1	Keyboard	1200.0
2	Mouse	800.0
3	Chair	2500.0
4	Desk	5500.0

View contents after updating Price of ProductID = 1 to 1300 via the view:

ProductID	ProductName	Price
1	Keyboard	1300.0
2	Mouse	800.0
3	Chair	2500.0
4	Desk	5500.0

Question 9: Stored procedure output by category

CALL GetProductsByCategory('Electronics');

ProductID	ProductName	Category	Price
1	Keyboard	Electronics	1300.0
2	Mouse	Electronics	800.0

CALL `GetProductsByCategory('Furniture')`:

ProductID	ProductName	Category	Price
3	Chair	Furniture	2500.0
4	Desk	Furniture	5500.0

Question 10: ProductArchive after AFTER DELETE trigger fires

Example: `DELETE FROM Products WHERE ProductID = 1`; the trigger copies the deleted row into ProductArchive with a DeletedAt timestamp.

ProductID	ProductName	Category	Price	DeletedAt
1	Keyboard	Electronics	1300.0	2025-01-01 10:15:32